

Mathematical Methods I: Euclidean Space

Inner Products, Norms, and Orthogonality

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1. Euclidean Space

The n -dimensional **Euclidean space**, denoted by $\mathbb{R}^n$ (or E^n), formalizes Cartesian geometry using real numbers. Its elements are points or vectors expressed as ordered n -tuples:

$$\mathbf{x} = (x_1, x_2, \dots, x_n), \quad x_i \in \mathbb{R}. \quad (1)$$

Vector addition and scalar multiplication are performed componentwise:

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_n + y_n), \quad \lambda \mathbf{x} = (\lambda x_1, \dots, \lambda x_n). \quad (2)$$

Thus, $\mathbb{R}^n$ is a real vector space. Its geometric structure emerges when it is equipped with an inner product, which allows us to define lengths, distances, and angles.

Definition 1.1 (Inner Product)

Let V be a real vector space. An **inner product** is a function that assigns a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ to each pair of vectors $\mathbf{u}, \mathbf{v} \in V$ and, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $k \in \mathbb{R}$, satisfies:

1. **Symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
2. **Additivity:** $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$.
3. **Homogeneity:** $\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$.
4. **Positive definiteness:** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$.

A real vector space equipped with an inner product is called a **real inner product space**.

2. Euclidean Inner Product

Example 2.1 (Dot Product in $\mathbb{R}^n$)

For $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$, define

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n = \sum_{i=1}^n u_i v_i. \quad (3)$$

This is the **Euclidean inner product**, also called the **standard inner product**. For example, if $\mathbf{u} = (1, 2, -1)$ and $\mathbf{v} = (3, 0, 4)$, then

$$\langle \mathbf{u}, \mathbf{v} \rangle = 1(3) + 2(0) - 1(4) = -1.$$

The axioms imply several useful identities:

$$\langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0, \quad (4)$$

$$\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle, \quad (5)$$

$$\langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle, \quad (6)$$

$$\langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle, \quad (7)$$

$$k \langle \mathbf{u}, \mathbf{v} \rangle = \langle k\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, k\mathbf{v} \rangle. \quad (8)$$

3. Norm, Distance, and Unit Vectors

Definition 3.1 (Induced Norm and Distance)

The **norm**, or length, of a vector $\mathbf{v} \in V$ is defined by

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}. \quad (9)$$

The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is the length of their difference:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}. \quad (10)$$

A vector whose norm equals one is called a **unit vector**.

In $\mathbb{R}^n$ with the standard inner product, these expressions become

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \dots + v_n^2}, \quad d(\mathbf{u}, \mathbf{v}) = \sqrt{\sum_{i=1}^n (u_i - v_i)^2}. \quad (11)$$

Theorem 3.2 (Basic Properties)

For $\mathbf{u}, \mathbf{v} \in V$ and $k \in \mathbb{R}$:

1. $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
2. $\|k\mathbf{v}\| = |k| \|\mathbf{v}\|$.
3. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$.
4. $d(\mathbf{u}, \mathbf{v}) \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{v}$.

If $\mathbf{v} \neq \mathbf{0}$, it can be normalized by dividing it by its length:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \|\hat{\mathbf{v}}\| = 1. \quad (12)$$

The Unit Sphere**Definition 3.3 (Unit Sphere)**

The **unit sphere** of an inner product space V is the set

$$S = \{\mathbf{u} \in V : \|\mathbf{u}\| = 1\}. \quad (13)$$

In $\mathbb{R}^2$, it is the unit circle; in $\mathbb{R}^3$, it is the surface of the sphere of radius one.

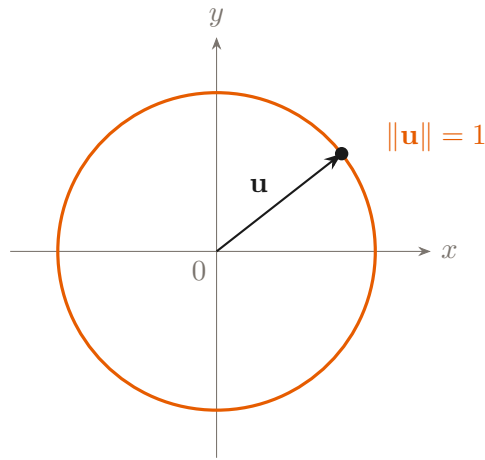


Figure 1: The unit circle is the unit sphere of $\mathbb{R}^2$.

4. Orthogonality and Angles

Definition 4.1 (Orthogonal Vectors)

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, when

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0. \quad (14)$$

The zero vector is orthogonal to every vector. For nonzero vectors, this condition corresponds geometrically to a right angle.

For $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$, the angle $\theta \in [0, \pi]$ between them is defined by

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (15)$$

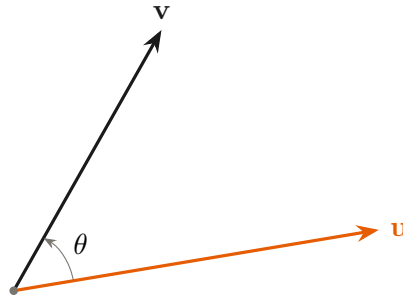


Figure 2: The inner product determines the angle between two nonzero vectors.

Theorem 4.2 (Generalized Pythagorean Theorem)

If $\mathbf{u} \perp \mathbf{v}$, then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2. \quad (16)$$

Indeed, expanding the squared norm makes the cross terms vanish:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2.$$

5. Fundamental Inequalities

Theorem 5.1 (Cauchy–Schwarz Inequality)

For any $\mathbf{u}, \mathbf{v}$ in a real inner product space,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|. \quad (17)$$

If both vectors are nonzero, equality holds precisely when one is a scalar multiple of the other. This inequality guarantees that the expression used to define $\cos \theta$ always lies in the interval $[-1, 1]$.

Theorem 5.2 (Triangle Inequality)

For $\mathbf{u}, \mathbf{v} \in V$,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|. \quad (18)$$

In terms of distance, for any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$,

$$d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v}). \quad (19)$$

Geometrically, the direct segment between two points is never longer than a path that passes through a third point.

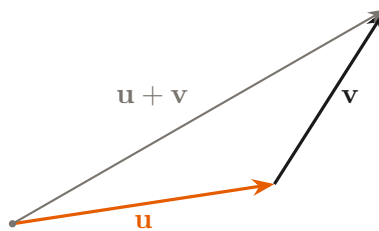


Figure 3: Geometric interpretation of the triangle inequality.

6. Comprehensive Example

Example 6.1

Let $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (2, -1)$. Then

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= 1(2) + 2(-1) = 0, \\ \|\mathbf{u}\| &= \sqrt{1^2 + 2^2} = \sqrt{5}, \\ \|\mathbf{v}\| &= \sqrt{2^2 + (-1)^2} = \sqrt{5}.\end{aligned}$$

Therefore, $\mathbf{u} \perp \mathbf{v}$. Furthermore,

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|(3, 1)\|^2 = 10 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2,$$

which verifies the generalized Pythagorean theorem. The normalized vectors are

$$\hat{\mathbf{u}} = \frac{1}{\sqrt{5}}(1, 2), \quad \hat{\mathbf{v}} = \frac{1}{\sqrt{5}}(2, -1).$$

Thus, they form a pair of orthogonal unit vectors.